

Rudra Pratap Singh

Full Stack Developer | AI Engineer

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Education

Indian Institute of Technology Roorkee, BS-MS in Mathematics and Computing Nov 2021 – May 2026

Experience

Lead Engineer (Full Stack) Milk. Inc June 2025 – Present

- Built the FastAPI + PostgreSQL backend for [ANSWER-Lite](#), a production hyperspectral data platform (exhibited at FOOMA JAPAN 2026) with SQLAlchemy 2.0, Alembic migrations, and Pydantic v2 across a bilingual multi-tenant deployment.
- Designed the hardware integration API for the IRODORI 18-band hyperspectral camera (410–940nm), letting it push spectral data directly to the backend with configurable per-device targets and labels.
- Set up async job processing with **Celery + Redis** for long-running spectral analysis tasks, decoupled from the FastAPI request cycle; maintained a GitHub Actions CI/CD pipeline for a Dockerized FastAPI + React stack on AWS Lightsail with staging/production separation and a zero-downtime instance migration (snapshot upgrade, static IP reassignment, DNS cutover).
- Built a Next.js 15 App Router SaaS app for Japanese landlords, deployed on Cloudflare Pages, with a three-context **Supabase** client (browser/server/middleware) and row-level security enforcing per-owner data isolation.
- Integrated **Stripe** multi-tier subscriptions (basic/standard/premium) and one-time payments with manual HMAC webhook verification; built a Google Gemini Pro OCR pipeline for AI document extraction with per-user monthly quota tracking via atomic Supabase RPC.
- Automated preprocessing of 500+ award photos using `face_crop_plus` and `birefnet-portrait`, achieving >98% accuracy in alignment and background consistency.

AI Engineer Intern DeeCogs Technologies Pvt. Ltd. February 2025 – August 2025

- Developed PhysioTrack, a Python library for real-time pose estimation using **MMPose** and **RTMLib**, calculating joint angles from video/webcam feeds to assess spinal mobility for physiotherapy use.
- Built an AI physiotherapy agent with Google ADK and WebSockets for real-time bidirectional patient communication, analyzing live video/audio and generating automated session feedback.

Projects

TruckNet Source Code | Deployed Link March 2025 – April 2025

- Built a load bidding platform where shippers post loads and truckers bid live, with the lowest bid winning.
- Implemented WebSocket-powered bidding rooms and JWT-authenticated RESTful APIs, with a SuperAdmin dashboard for approvals and oversight.

Technical Skills

- Languages:** Python (Pandas, NumPy), C++ , JavaScript/TypeScript, SQL
- Frameworks:** FastAPI, React, Next.js, Node.js, Express, SQLAlchemy, Celery, PyTorch
- Tools:** Git, Docker, AWS, PostgreSQL, Redis, Supabase, MongoDB, Linux, GitHub Actions
- AI/ML:** LangChain, Google Gemini, Google ADK, MMPose, RTMLib, FAISS

Achievements

- All India Rank 4563, JEE Advanced 2021:** Achieved AIR 4563 in JEE Advanced 2021 out of 1,51,000 candidates.
- Specialist on Codeforces:** Maximum Rating of 1551 on Codeforces (Handle - chipi-chipi_chapa-chapa).
- National Finalist, Code With Cisco:** Ranked in the top 0.25% of 30,000+ participants nationwide.

Positions of Responsibility

Cup Captain Nov 2023 – Jan 2024

- Led a contingent of 50+ members across Parliamentary Debates, Standup Comedy and Improv Comedy.
- Contributed to securing overall Second Runner Up in Speaking Arts across all IIT's in Inter-IIT Cult Meet 2023.